

An illustration of a team of seven people in a modern office setting, gathered around a whiteboard. The team consists of five men and two women, all dressed in casual business attire. They are looking at the whiteboard, which is covered with handwritten notes and diagrams. One man is pointing at a note on the board. The background shows large windows with a city view and indoor plants. The overall color palette is muted, with a blue and purple tint.

Sprint 6

Team 1

20 July 2026

Sprint W28 Review

Pipeline Convergence • Week 28 • Agile Quant Trading & Data Engineering

A full-team sprint review covering pipeline integration milestones, market regime forecasting, scrum health, technical analysis, and model calibration. Every agent delivered — from the Almanac to the Delta Engine.

SPRINT 28

INTERNAL REVIEW

CONFIDENTIAL



Sprint W28 Goal: Pipeline Convergence

Sprint 28 delivered the single most important integration milestone of the project cycle: the full end-to-end pipeline ran autonomously, producing predictions and fetching live data without manual intervention. Three discrete engineering achievements converged to make this possible.

Backend × Frontend Integration

The backend web server was successfully integrated with the frontend UI. Both systems now share a unified data contract, enabling real-time display of pipeline outputs directly in the interface.

Delta Engine Deployed

The Delta Engine — responsible for comparing locked predictions against actuals — was merged into the main pipeline. This closes the calibration loop and enables dynamic weight adjustment between sprints.

Macro Graph Generation Live

Macro Graph Generation was deployed end-to-end, automatically fetching macroeconomic data and rendering structured visual outputs. No manual data pulls required.

- ✅ Definition of Done Met: The full pipeline ran successfully end-to-end, producing predictions and fetching data autonomously — the primary Sprint 28 acceptance criterion.

Final Prediction & Human Override

Regime Signal

Neutral-Bullish

Medium Confidence

Bullish technical structure faces a direct collision with heavy Federal Reserve event risk. The regime is constructive but fragile — a single hawkish data point could flip the bias intraday.

Forecast Ranges — Sprint W28

SPX · FLAT-UP

Range: **-0.5% to +1.2%** | Technicals are constructive. Price above both EMAs but resistance at 7,578 is a near-term ceiling. Upside gated by Fed outcome.

NDX · UP

Range: **-0.2% to +2.0%** | Strongest risk-adjusted setup of the three. Mega-cap tech momentum is intact. Model consensus skews bullish assuming no hawkish surprise.

IWM · FLAT

Range: **-1.5% to +0.5%** | Small caps remain structurally weak. Rate sensitivity is high and IWM has slipped below the 8 EMA. Downside risk is asymmetric.

Scrum Health: Triumphs & Blockers

Sprint 28 delivered its headline goal but was not without friction. The team demonstrated strong engineering execution while exposing structural vulnerabilities in API dependency management and communication discipline.

✓ What Worked

Sprint Goals Confirmed Clean

All primary acceptance criteria were met and signed off without scope renegotiation. The Definition of Done was honored in full — a significant improvement over prior sprints.

Backend × Frontend Unity

After multiple sprints of parallel development, the backend and frontend systems were finally integrated into a single cohesive pipeline. This unblocks downstream feature delivery for Sprint 29 and beyond.

● Major Blocker

OpenRouter API Rate Limits

Unexpected rate limiting from the OpenRouter API caused mid-sprint pipeline failures that required unplanned troubleshooting cycles. This consumed engineering bandwidth that was scoped for feature work and delayed integration testing by an estimated one full day.

Root Cause

The pipeline lacked a fallback or backup API key rotation strategy. Single-provider dependency created a single point of failure under load. This must be resolved before Sprint 29 live runs.

Team Action Items & Improvements

Sprint 28 surfaced two systemic issues that need structural remediation — not just acknowledgment. Uneven participation and fragmented communication are team-level risks that compound over time and erode sprint velocity. The following action items are binding for Sprint 29.

1 Prepare Backup API Keys for Fault Tolerance

All critical external API dependencies must have at least one backup key or provider fallback configured before the Sprint 29 kickoff. Responsibility: Engineering leads. Verification: Pipeline smoke test on Day 1 of Sprint 29.

2 Mandate Daily Public Progress Updates

All team members must post a brief daily status update in the main shared channel — not in DMs. Aeron, Henry, and Saiko reported zero measurable progress in Sprint 28. This pattern is unacceptable and will be tracked explicitly in Sprint 29 retrospectives.

3 Complete Work Earlier to Buffer Integration Testing

Integration testing must begin no later than Day 7 of each sprint. Work delivered in the final 24 hours of a sprint cannot be properly tested, reviewed, or merged without risk. Earlier delivery is a team commitment, not a suggestion.

 Zero-progress contributors in Sprint 28: Aeron, Henry, Saiko. This will be reviewed explicitly in the Sprint 29 kickoff. Recurring non-participation will escalate to the Product Owner.

Almanac Agent: Seasonal Crosswinds

Cycle Context

We are currently positioned inside the **Midterm Election Year Q2–Q3 "Weak Spot"** — a historically documented period of subdued or choppy equity performance in midterm election cycles. Seasonal tailwinds that supported earlier quarters have largely dissipated.

Overall Bias

Mixed · Medium Confidence

Cross-sector divergence makes a unified directional call unreliable. Sector selection — not broad index exposure — is the alpha driver in this environment.

Sector Divergence Matrix

LONG: Technology (XLK)

XLK remains in a powerful and well-established long window per seasonal modeling. Historical data across midterm election cycles confirms sustained tech outperformance in this specific Q2–Q3 window. Maintain long bias until seasonal model signals rotation.

SHORT: Financials, Materials, Energy

XLF, Materials, and Energy sectors are all flagged under active SHORT windows in the Almanac model. Rising rates structurally pressure financials, while commodity sectors face demand-side headwinds from a slowing growth narrative. Reduce or hedge exposure.

Macro Agent: The Binary Risk Catalyst

The macroeconomic environment in Week 28 was defined by a single dominant event: the US Federal Reserve Interest Rate Decision. With a 97.4% probability of a hold priced in by futures markets, the base case was widely expected — but the *tone* of the statement and press conference carried enormous tail-risk potential for equity markets already trading near resistance.



Fed Rate Decision — 97.4% Hold

The binary risk is not in the decision itself but in the Fed's forward guidance. Any language more hawkish than expected — extended pause, upward dot revision, or pushback on rate cut timelines — would shock a market positioned for eventual easing. Equities are trading on Fed optionality, and any reduction in that optionality is a direct negative catalyst.



10-Year Treasury Yield: 4.56%

The 10-Year yield crept higher to 4.56% during the week — a level that mechanically pressures equity multiples, particularly in rate-sensitive sectors. Growth stocks, which dominate NDX, are especially vulnerable to yield expansion. The move signals that the bond market is not fully aligned with the equity market's bullish pricing.



WTI Crude Oil: +4.18% Surge

A sharp +4.18% move in WTI Crude added inflationary pressure to the macro picture. Energy price spikes historically complicate Fed policy messaging and can stoke renewed inflation fears. If crude sustains this level, it creates a secondary headwind for rate cut expectations into Q4.

⊗ Macro Bias: Binary-Risk. Equities are extremely sensitive to any hawkish surprise. A single sentence in the Fed statement could override all bullish technical signals.

Technical Structure: Large Caps Dominate

The technical picture is bifurcated and concentration-dependent. SPX and NDX are exhibiting textbook momentum structure — both indices trading comfortably above their 8 EMA and 21 EMA, placing them firmly in Zone 1 (Bullish). IWM tells a different story entirely, having slipped below its 8 EMA and entering Zone 2 (Neutral), a signal that small caps are absorbing the full weight of elevated interest rates.

SPX — Zone 1 Bullish

Trading at 7,571.01, comfortably above both the 8 EMA (~7,510) and 21 EMA (~7,472). The trend structure is clean and intact. Key support at 7,257. Resistance at 7,578 — the index is testing this level in real time. A clean break above 7,578 opens the next leg higher.

NDX — Zone 1 Bullish

At 29,818.69, NDX has the strongest technical setup of the three. The 8 EMA (~29,604) and 21 EMA (~29,593) are nearly converged and providing a tight, rising support band. Upside target: 30,643. Key support at 28,548. Mega-cap tech momentum is the primary driver of this structure.

IWM — Zone 2 Neutral

At 295.89, IWM has slipped below the 8 EMA (~296.64) — a bearish intraday signal. The 21 EMA (~294.76) is providing immediate floor support. Small caps are structurally lagging. High rates disproportionately impact smaller companies with floating-rate debt exposure. Do not chase long exposure here.

 **Concentration Risk:** The rally is highly concentrated in mega-cap technology. Breadth is narrow. If NDX tech names rotate or take profit, SPX support at 7,472 becomes the first critical test.

Technical Support & Resistance Matrix

The following key levels table represents the definitive technical reference for Sprint W28. These levels were locked at the time of prediction and serve as the ground truth against which the Delta Engine measures directional accuracy.

Index	Current Close	8 EMA	21 EMA	Trend Bias	Key Support	Key Resistance	Zone
SPX	7,571.01	~7,510	~7,472	Bullish	7,257	7,578	Zone 1
NDX	29,818.69	~29,604	~29,593	Bullish	28,548	30,643	Zone 1
IWM	295.89	~296.64	~294.76	Neutral	283.40	302.72	Zone 2

IWM is the outlier — the only index trading *below* its 8 EMA, placing it in a structurally different regime from SPX and NDX. Rate sensitivity is the primary explanatory variable. Monitor the 294.76 level (21 EMA) as the critical floor; a breach would shift IWM into Zone 3 (Bearish) and warrant a defensive posture on small-cap exposure.

Week 08 Actual Outcomes

Index Performance — Ground Truth

+1.67%

Nasdaq (NDX)

Led all major indices. Mega-cap tech drove outsized returns. Model directional call: correct.

+1.17%

S&P 500 (SPX)

Solid follow-through above key resistance at 7,578. Model directional call: correct.

-0.57%

Russell 2000 (IWM)

Small caps lagged as predicted. Neutral Zone 2 call validated. Downside within forecast range.

Sector Breadth & Volatility

Energy: +2.87%

Strongest sector performer despite Almanac SHORT signal. WTI crude surge (+4.18%) drove energy equity outperformance — a model miss worth calibrating.

Technology: +2.83%

Second-best performer. Aligned precisely with the Almanac LONG signal on XLK and the Technical Zone 1 bullish call. High conviction, high accuracy.

VIX: 15.28 (−5.39%)

Volatility compressed significantly. Market complacency is elevated — historically a precursor to sharp moves when a catalyst (Fed) finally resolves. Watch for VIX spike post-Fed.

Human Score: Overriding AI Optimism

The Human Score layer exists to apply qualitative judgment that quantitative models systematically underweight. In Sprint W28, the Human Override introduced a critical penalty that moderated the AI ensemble's net bullish bias — a correction that proved directionally sound given actual outcomes.

Human Score Summary

+2

Final Score: **Neutral-Bullish**

The raw AI ensemble skewed higher. The Human Override applied a structured penalty to bring the net score into a more defensible range given the macro environment.

Wild Card Penalty: -2 Applied

AI Underestimated Fed Event Risk

The quantitative models assigned insufficient weight to the binary risk posed by the Federal Reserve decision. A 97.4% hold probability does not neutralize the tail risk from hawkish guidance language — and that tail risk was large enough to warrant a meaningful penalty on the bullish signal.

Tech Carrying Market Near Resistance

NDX approaching 30,643 resistance while breadth remains narrow is a structurally fragile setup. The Human Score penalized overconfidence in momentum continuation at resistance levels — a judgment that pure price-action models tend to miss.

Profit-Taking Risk at Extremes

A hawkish Fed tone would not merely create a modest pullback — it could trigger cascading profit-taking in an overbought tech sector. This asymmetric downside risk was explicitly captured in the Human Score penalty and not adequately represented in the AI ensemble output.

LLM Operator: Multi-Model Synthesis

Sprint W28 marked a significant engineering milestone for the LLM Operator function: four top-tier large language models were successfully queried programmatically via API in a single unified pipeline run. The outputs were aggregated into a structured synthesis that surfaces both consensus signals and model-specific divergences — the raw material for the final regime call.

Engineering Win

Nemotron 3, Gemma 4, Laguna M.1, and Tencent Hy3 were all queried automatically via the OpenRouter API. No manual prompting required. This is the first sprint where multi-LLM synthesis ran entirely programmatically — a foundational capability for the autonomous prediction engine.

Consensus Narrative

Neutral-Bullish across all four models. While individual range estimates differ, all models agree on directionality: constructive near-term bias with meaningful upside risk in NDX and limited downside in SPX assuming no hawkish Fed shock.

Core Conflict: Tug-of-War

Every model identified the same fundamental tension: **excellent price structure** (SPX and NDX above both EMAs, clean trend) vs. **dangerous macro limits** (Fed rate policy, rising 10-Year yield at 4.56%, crude oil spike). Resolution of this tension is the single most important variable for the next sprint.

LLM Comparison Matrix

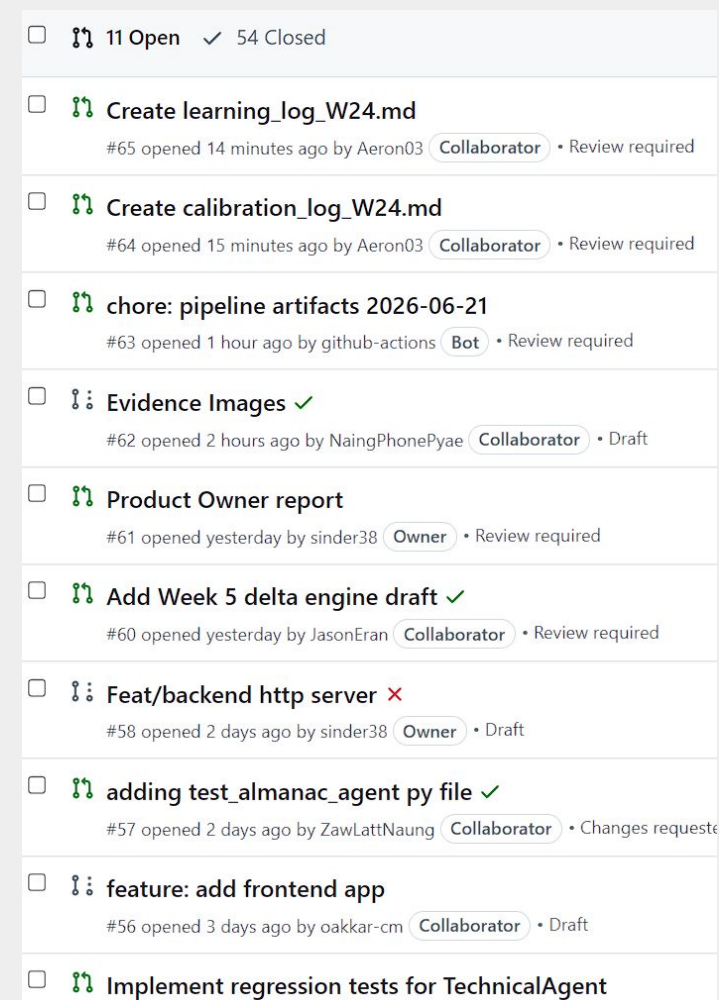
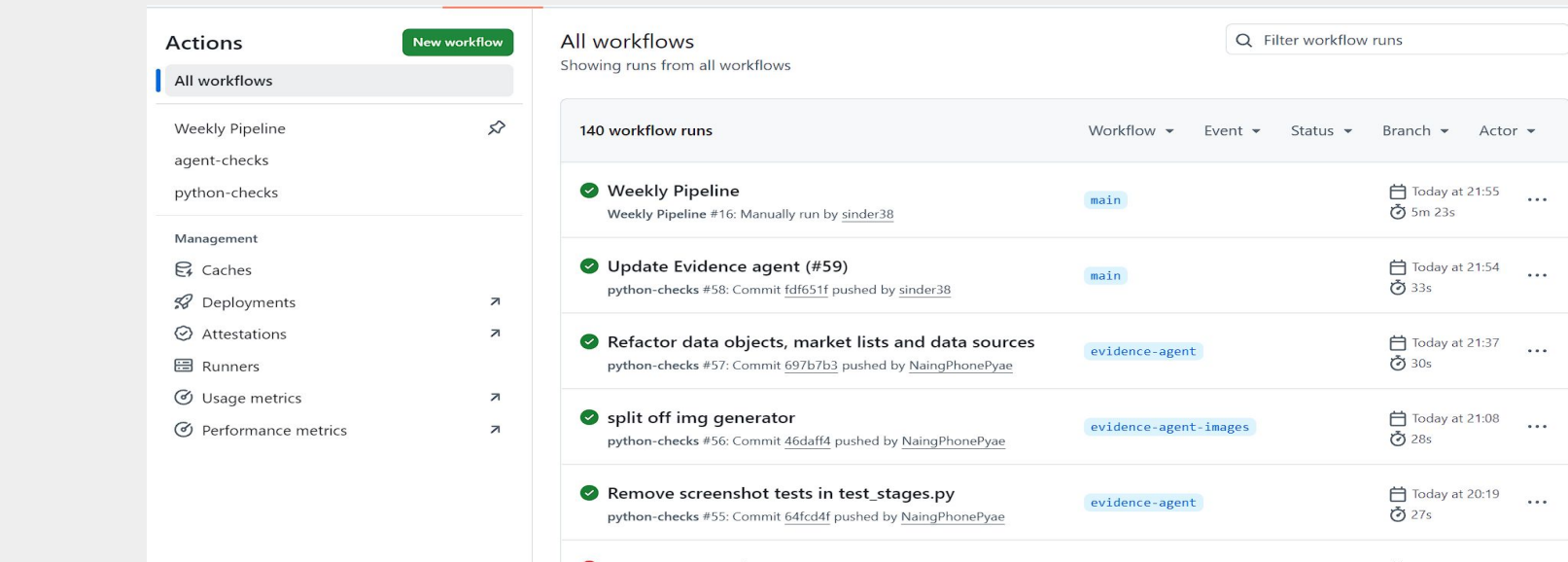
The following table captures the raw output from each of the four LLMs queried in Sprint W28. Divergences in range estimates and stated contradictions are the primary inputs to the Human Score layer and should inform model weighting adjustments in the Delta Engine for Sprint 29.

LLM Model	Regime	Confidence	SPX Est.	NDX Est.	Key Contradiction
Nemotron 3	Bullish	Medium	-4.1% to +0.1%	-4.3% to +2.8%	Dovish Fed surprise needed to sustain upside
Gemma 4	Bullish	Medium	-0.5% to +1.2%	-0.2% to +2.0%	Rising 10Y Yield creating multiple compression pressure
Laguna M.1	Mixed	Medium	-4.1% to +0.1%	-4.2% to +2.7%	Upside capped by resistance; broader macro deterioration
Tencent Hy3	Neutral	Medium	-4.1% to +0.1%	-4.3% to +2.8%	Mixed macro vs. isolated Tech strength — breadth missing

Notable divergence: **Gemma 4** produced the tightest and most optimistic SPX range (-0.5% to +1.2%), while the remaining three models produced a much wider band (-4.1% to +0.1%) — reflecting higher uncertainty around the downside scenario. The Delta Engine will weight these discrepancies in its next calibration cycle based on actual outcome matching.

GitHub Repository

- **System Integration:** Successfully connected all isolated Agents into a unified web server pipeline.
- **High Automation Achieved:** High level of automation achieved, eliminating manual friction for data provisioning. PRs merged and reviewed systematically.
- **New job** for enforcing code correctness
- **5 LLM models**
- **Tech Debt Eradication:** Conducted comprehensive code quality reviews and removed critical technical debt to stabilize the pipeline.

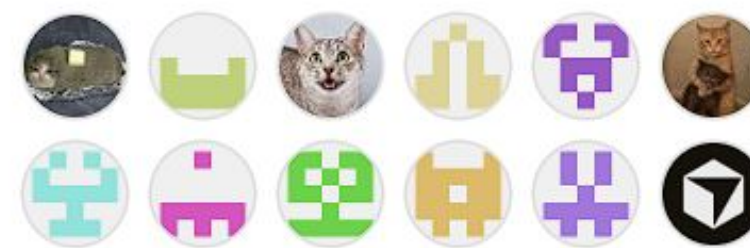


Deployments 6

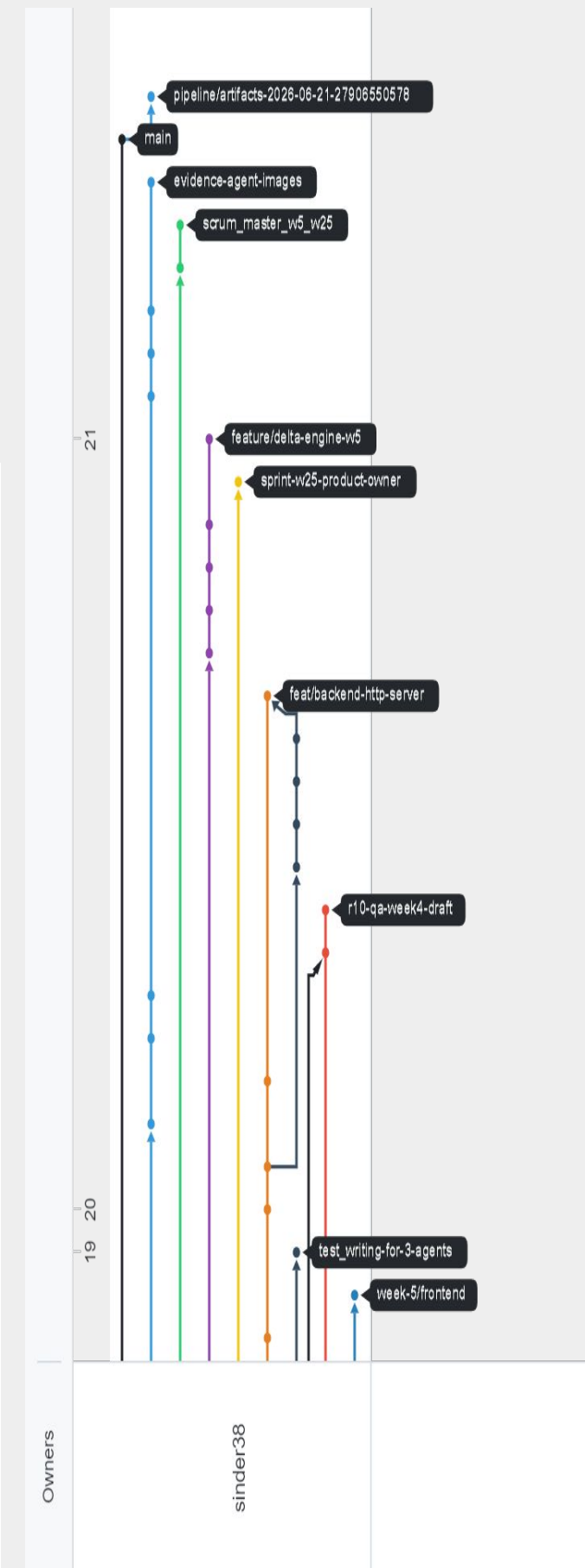
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[+ 5 deployments](#)

Contributors 12



Languages



Calibration: Closing the Loop

Delta Engine — Sprint W28 Milestone

The Delta Engine represents the most strategically important infrastructure delivered in Sprint 28. For the first time, the system can automatically close the prediction loop — comparing what was forecast against what actually happened, without human intervention in the calculation process.

01

Lock Predictions at Sprint Start

All agent forecasts (regime, range, direction) are locked at the beginning of each sprint and stored immutably as the prediction record.

02

Ingest Actual Market Results

At sprint close, the Data Collector feeds actual index performance, sector returns, and volatility readings into the pipeline automatically.

03

Compute Accuracy Metrics

The Delta Engine calculates direction accuracy, range accuracy, and error percentage for each model and each agent across all three indices.

04

Dynamically Adjust Weights

Models that consistently outperform receive increased weight in the ensemble. Underperforming models are penalized. Weights propagate automatically into Sprint 29 predictions.

Impact

Direction Accuracy

Was the directional call (UP / DOWN / FLAT) correct for each index?

Range Accuracy

Did the actual return fall within the predicted forecast band?

Error %

Quantifies the magnitude of miss when the forecast range was incorrect — used to penalize overconfident or underconfident models.

- ✔ Sprint 28 Result: Delta Engine is live and connected. Sprint 29 will be the first sprint to benefit from automated weight adjustment based on W28 actuals.

Thank You

Do you have any questions?